

WASP-36b

R-2175

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_140225 · created 2026-08-13T00:57:35+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456713.63641 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.047
Transit depth (Rp/Rs)^2	2.74 +/- 1.55 [%]
Orbital Inclination (inc)	82.83 +/- 2.9
Airmass coefficient 1 (a1)	0.6 +/- 0.37
Airmass coefficient 2 (a2)	0.34 +/- 0.36
Scatter in the residuals of the lightcurve fit is	61.99 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	3.0
Transit Duration (day)	0.061 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.047 vs 0.1368 (deviation 0.6σ)
Duration fit vs published	0.061 d vs 0.0758 d (deviation 0.71σ)
Mid-time O–C	-2.2 min
Depth SNR	3.5
Residual scatter	61.99 %

Light curve

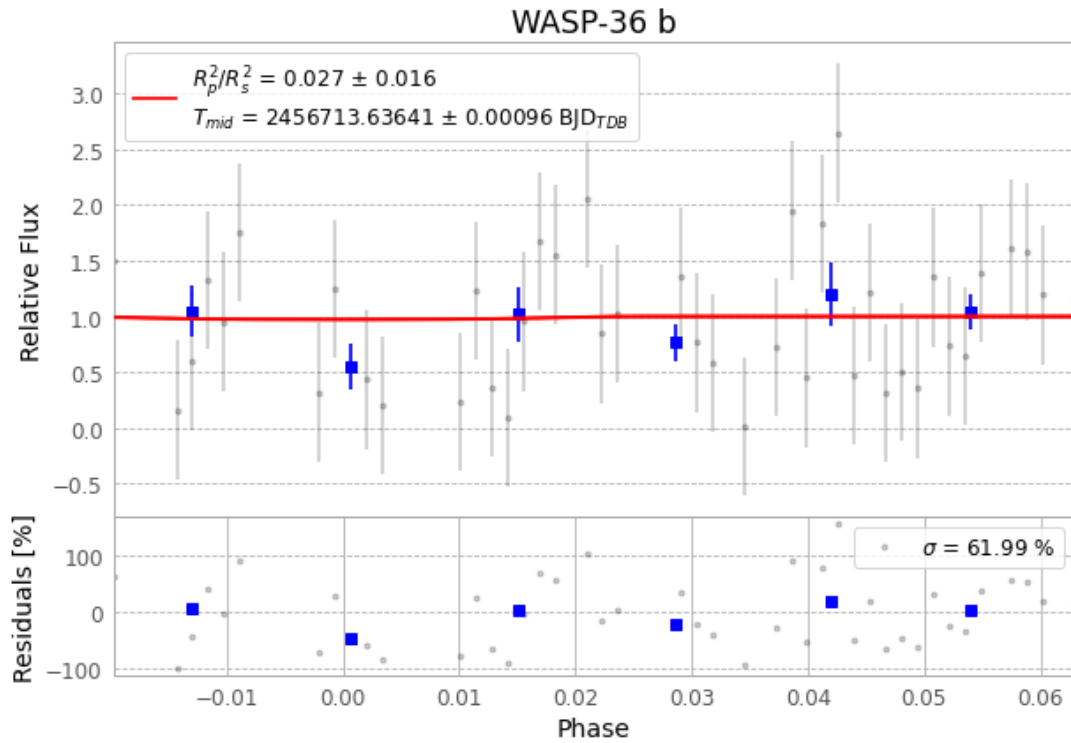


Figure 1 — FinalLightCurve_WASP-36 b_2014-02-24.png (SHA-256 b4a9c2fc4aaa3455...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [309, 346], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7dd60251f5844e4e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

64 FITS frames (42 MB), WASP-36140225021515.FITS → WASP-36140225053013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-140225.log (6dc58e2b6943...), FinalLightCurve_WASP-36 b_2014-02-24.png (b4a9c2fc4aaa...), FinalLightCurve_WASP-36 b_2014-02-24.csv (ee7d29285d11...)

Compute resources

Wall time 138.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 00:57Z.